

Coulomb excitation **1969Ha17,1977Sc36**

1969Ha17: 51-61-MeV $^{35}\text{Cl}^{5,6+}$ beams of 80 nA were produced from the Chalk River MP tandem accelerator and bombarded 9.6-mg/cm² carbon and 5-mg/cm² ^{24}Mg . Deexcitation γ rays were detected using two 40-cm³ Ge(Li) detectors at 125° and at 0° or 90°. Measured E_γ , I_γ , $\gamma(\theta)$ anisotropy, γ -ray yields. Deduced levels, mixing ratio, transition strengths, $T_{1/2}$ for 2 levels using the Doppler Shift Attenuation Method (DSAM).

1977Sc36: 53-56-MeV $^{35}\text{Cl}^{6+}$ beams of 60 nA were produced from a single MP tandem Van de Graaff accelerator at the three-stage facility of Brookhaven National Laboratory. Targets were ^{23}Na and ^{27}Al . Deexcitation γ rays were detected using a 55-cm³ coaxial Ge(Li) detector at 0°. Measured E_γ , $I_\gamma(\theta)$, γ -ray yields. Deduced levels, mixing ratios, transition strengths, $T_{1/2}$ for the levels of 1220, 1762 and 2650 keV using the Doppler Shift Attenuation Method (DSAM).

 ^{35}Cl Levels

Values of $B(E2)\uparrow$ from **1969Ha17** and **1977Sc36** are from absolute cross-section measurements.

$E(\text{level})^\dagger$	$J^\pi^\dagger$	$T_{1/2}$	Comments
0	$3/2^+$		
1219	$1/2^+$	0.150 ps 20	$B(E2)\uparrow=0.00076\ 7$ $T_{1/2}$: Lifetime $\tau=0.216$ ps 28 from weighted average of $\tau=0.16$ ps 5 (1969Ha17 , DSAM) and $\tau=0.230$ ps 25 (1977Sc36 , DSAM). $B(E2)\uparrow$: weighted average of 0.00081 9 (1969Ha17) and 0.00073 7 (1977Sc36). $B(E2)\uparrow=0.0106\ 7$
1763	$5/2^+$	0.42 ps 4	$T_{1/2}$: Lifetime $\tau=0.61$ ps 5 from weighted average of $\tau=0.57$ ps 7 (1969Ha17 , DSAM) and $\tau=0.63$ ps 5 (1977Sc36 , DSAM). $B(E2)\uparrow$: weighted average of 0.0117 9 (1969Ha17) and 0.0102 5 (1977Sc36). $B(E2)\uparrow=0.0029\ 7$ (1977Sc36)
2646	$7/2^+$	0.18 ps 6	$T_{1/2}$: Lifetime $\tau=0.26$ ps 9 (1977Sc36 , DSAM).

[†] From the Adopted Levels. Energies are rounded values.

 $\gamma(^{35}\text{Cl})$

$E_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [†]	δ	Comments
1219	1219	$1/2^+$	0	$3/2^+$	M1+E2	0.093 10	δ : deduced by the evaluators from adopted $B(E2)\uparrow=0.00076\ 7$ in this dataset and $T_{1/2}=0.119$ ps 14 from Adopted Levels. Other: 0.106 7 deduced by 1977Sc36 based on their measured $B(E2)\uparrow$ and lifetime.
1763	1763	$5/2^+$	0	$3/2^+$	M1+E2	-2.95 45	δ : from $\gamma(\theta)$ anisotropy in 1969Ha17 . Other: $1.8 +10-4$; deduced by the evaluators from adopted $B(E2)\uparrow=0.0106\ 7$ in this dataset and $T_{1/2}=0.36$ ps 3 from Adopted Levels; $-2.7\ 7$ from 1977Sc36 deduced from their measured $B(E2)\uparrow$ and lifetime.
2646	2646	$7/2^+$	0	$3/2^+$	E2		

[†] From Adopted Gammas. Energies are rounded values.

Coulomb excitation 1969Ha17,1977Sc36Level Scheme